Implied Volatility Analysis of AAPL Options Chains

A Quantitative Time-Series Study

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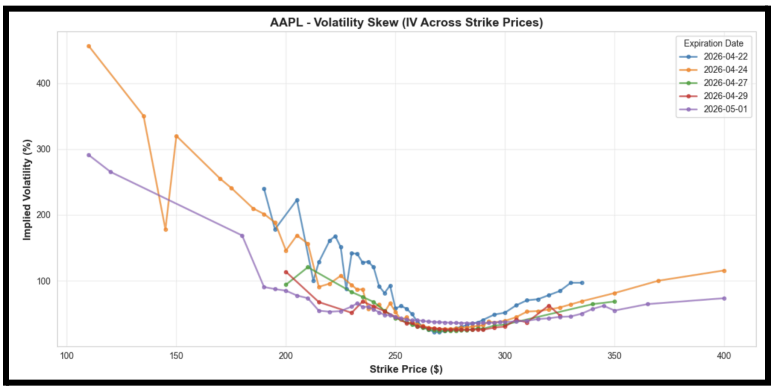
INTRODUCTION

This final displays a quantitative analysis of implied volatility (IV) in AAPL (Apple Inc.) options markets. This study examines options chain data collected on April 20, 2026, analyzing volatility patterns across different strike prices and expiration dates. The analysis aims to help understand market expectations for future volatility, investor risk sentiment, and trading activity distribution.

DATASET OVERVIEW

The dataset covers 409 call option contracts across 10 expiration dates ranging from April 22 to May 1, 2026, with strike prices from $5 to $450 and implied volatility ranging from 0% to 632%. After removing contracts with missing or invalid IV values, all 409 contracts were able to be used for analysis.

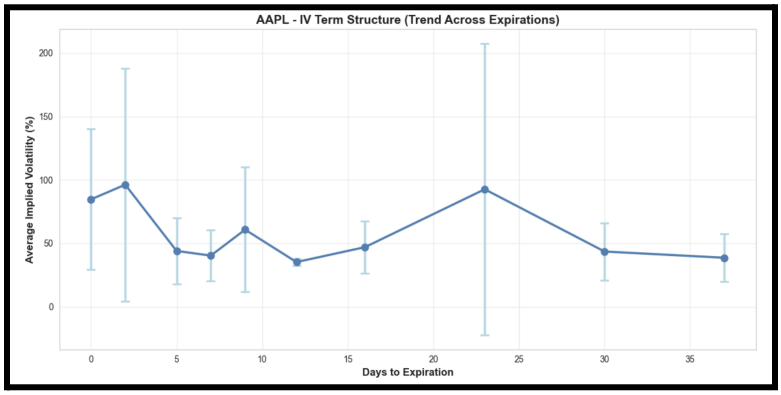
FINDING 1: VOLATILITY SKEW PATTERN



The volatility skew chart shows how IV changes as we move from lower strikes (out-of-the-money puts) through at-the-money (ATM) to higher strikes (out-of-the-money calls).

The chart exhibits a pronounced negative skew (AKA the 'volatility smirk'), where implied volatility is significantly higher for lower out-of-the-money strike prices and gradually decreases as strikes approach and exceed ATM. The market prices significantly higher risk for downside moves — OTM Put IV (128.45%) is substantially higher than OTM Call IV (43.75%) — meaning there's investor demand for downside protection, which is normal in risk-off environments.

FINDING 2: VOLATILITY TERM STRUCTURE

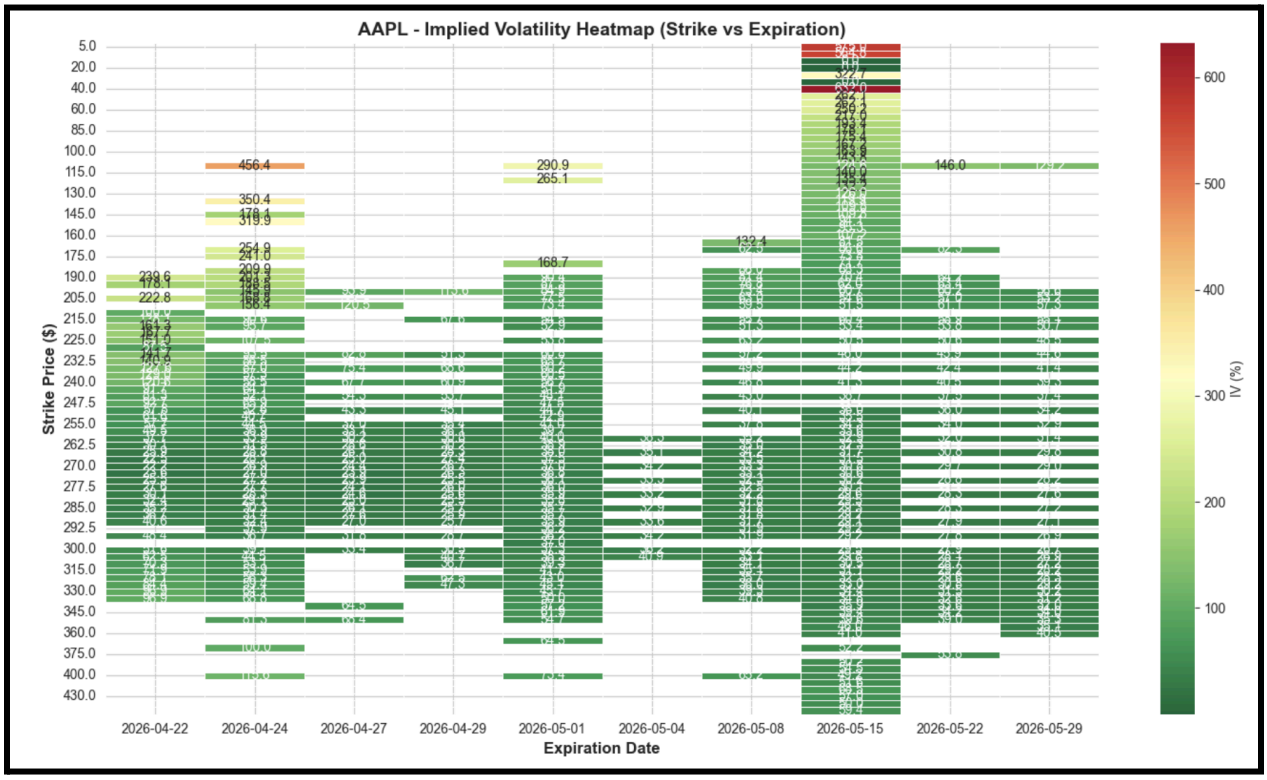


The term structure chart shows how IV changes across different expiration dates. The x-axis represents days until expiration, the y-axis shows average implied volatility, and the error bars show the range of volatility around each expiration.

The term structure slopes loosely downward, with IV very high at near-term expirations (84.63% at 0

DTE) and declining to approximately 38.60% for options 30+ days out. The wide error bars on near-term contracts suggest high variability in short-dated IV, which stabilizes further out. This structure usually signals an imminent catalyst, for example an earnings announcement, after which the market expects volatility to cool off over time.

FINDING 3: OVERALL VOLATILITY SURFACE



The heatmap above provides a comprehensive view of the entire IV picture. Rows represent strike prices, columns represent expiration dates, and colors represent IV levels (red = high, green = low).

The highest IV concentrations appear at lower strikes (around $40) and nearest expirations (April 22–24), while green zones cluster near ATM strikes (~$170) and longer-dated expirations. The figure combines a volatility smile across strikes and a descending term structure, revealing where the market thinks the greatest uncertainty is. This variation across the surface allows us to identify relatively cheap or expensive options in the entire options chain for NYSE: AAPL.

DESCRIPTIVE STATISTICS SUMMARY

Metric	Value	Metric	Value

Mean IV	66.71%	Minimum IV	0.00%
Median IV	41.00%	Maximum IV	632.03%
Std Deviation	71.92%	IV Range	632.03%

Expiration-Specific Insights:

1. 2026-04-24 (2 DTE): 96.19% ← Highest IV
2. 2026-05-15 (23 DTE): 92.58%
3. 2026-04-22 (0 DTE): 84.63%

Strike Concentration:

Highest IV Strike: $40 at 632.03%
Lowest IV Strike: $15 at 0.00%

Options trading is most concentrated at the $280 strike price expiring May 15 (41,182 contracts of open interest), so we can identify this as the primary zone of where larger institutions are going to be hedging.

To quantify the volatility skew, I put the options into three strike "zones": OTM puts (bottom 30th percentile), ATM (30th–70th), and OTM calls (top 30th), and average IV was computed per zone. For the term structure, contracts were organized by DTE (≤7, 8–30, 30+) and average IV calculated per group. These segmentation-based comparisons were the primary statistical framework for the findings above.

SUMMARY

To recap, this analysis examined implied volatility dynamics in AAPL options chains collected on April 21, 2026. Using three core visualizations (skew, term structure, heatmap) and statistical analysis, we identified:

- A pronounced negative skew where downside protection (OTM Puts) were priced at a significant premium.
- An inverted term structure that signaled high near-term uncertainty, but eventually normalizes over 30+ days.
- High liquidity clustering around the $280 strike, identifying the primary zone of institutional activity.

KEY TAKEAWAY

We can gather that the options market structure reveals pricing in increased risk for downside moves while expecting volatility to smooth out in the medium term. Professional traders are most active around the $280 strike and near-term expirations as well. This snapshot of the options market provides valuable information about market expectations as well as risk sentiment. Future analysis would benefit from multiple snapshots and analysis like this over time plus integration with OHLCV and positioning data.